

P7: Member Functions

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Overview

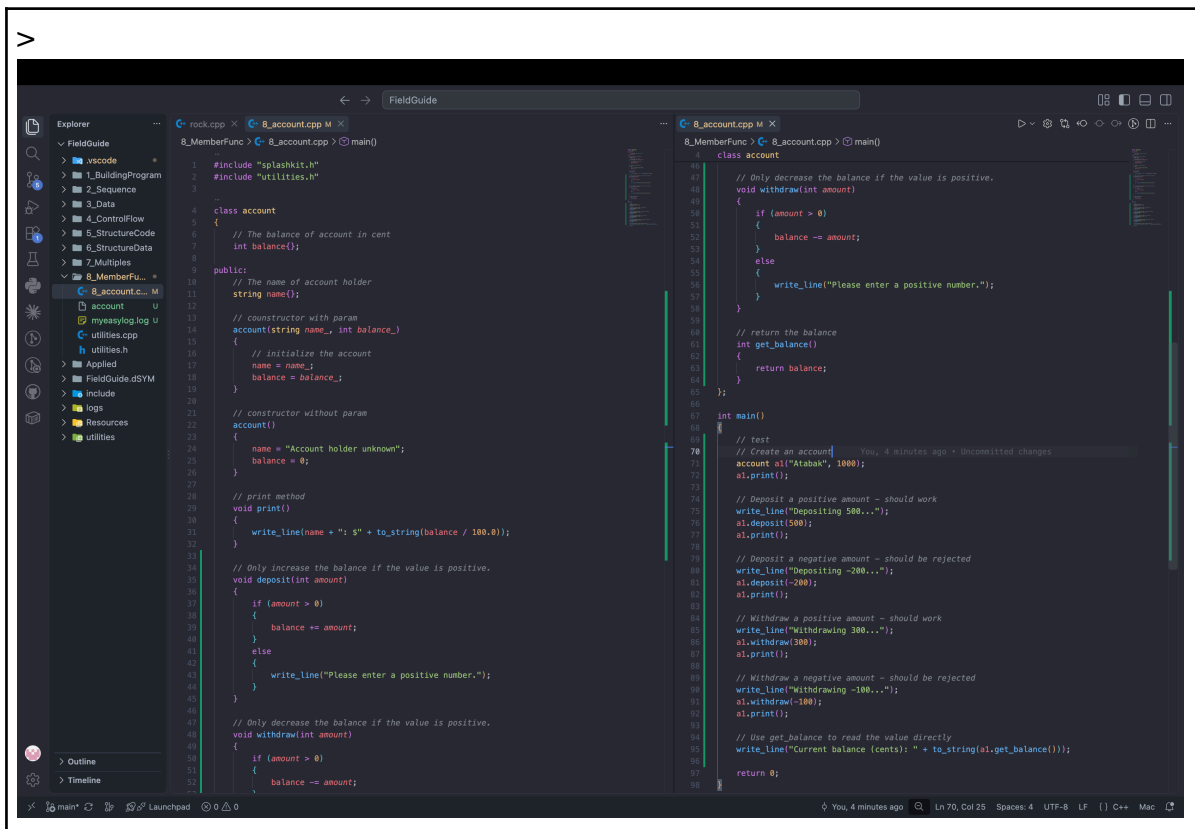
This task is an opportunity to explore and use language mechanisms that allow you to combine data and functionality within entities.

Tip: Systems thinking

Work to develop your ability to see how concepts come together as a system. The parts are not independent of each other. Understanding each part in isolation is a good start, but make sure to follow this up by seriously focusing on seeing how the parts come together. When you can see this, you can start to think about how to build and manipulate these systems, getting them to achieve what you want.

Do the following:

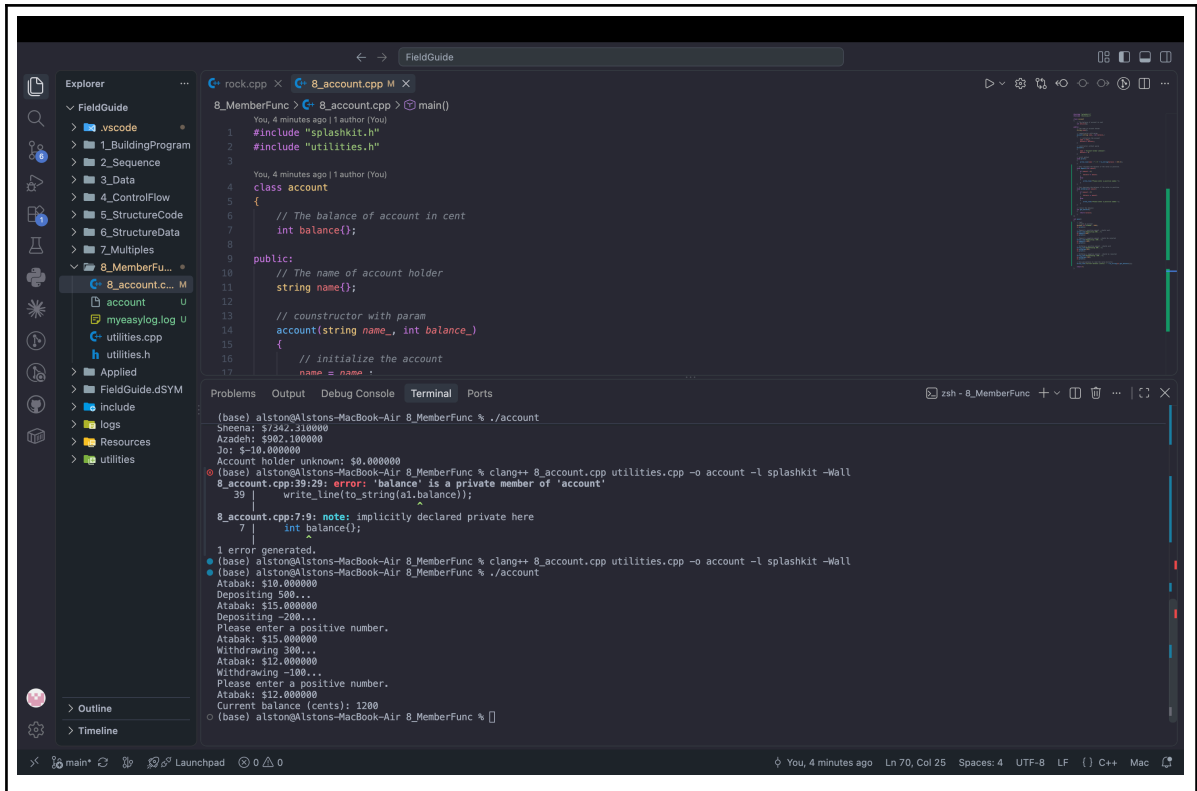
- Everything you need is in the Member Functions chapter of our [Programmer's Field Guide](#). The [Intro Tours](#) walk through the concepts, and should give you a good start.
- Add a screenshot showing you were able to create and work with member functions. You can use the [account class and objects tour](#) to achieve this.



```

1  #include "splashkit.h"
2  #include "utilities.h"
3
4  class account
5  {
6      // The balance of account in cent
7      int balance_;
8
9      public:
10         // The name of account holder
11         string name_;
12
13         // constructor with param
14         account(string name_, int balance_)
15         {
16             // initialize the account
17             name = name_;
18             balance = balance_;
19         }
20
21         // constructor without param
22         account()
23         {
24             name = "Account holder unknown";
25             balance = 0;
26         }
27
28         // print method
29         void print()
30         {
31             write_line(name + ": $" + to_string(balance / 100.0));
32         }
33
34         // Only increase the balance if the value is positive.
35         void deposit(int amount)
36         {
37             if (amount > 0)
38             {
39                 balance += amount;
40             }
41             else
42             {
43                 write_line("Please enter a positive number.");
44             }
45         }
46
47         // Only decrease the balance if the value is positive.
48         void withdraw(int amount)
49         {
50             if (amount > 0)
51             {
52                 balance -= amount;
53             }
54         }
55
56         // Only decrease the balance if the value is positive.
57         void withdraw(int amount)
58         {
59             if (amount > 0)
60             {
61                 balance -= amount;
62             }
63         }
64
65         // return the balance
66         int get_balance()
67         {
68             return balance;
69         }
70     };
71
72 int main()
73 {
74     // test
75     // Create an account
76     account al("Atabak", 1000);
77     al.print();
78
79     // Deposit a positive amount - should work
80     write_line("Depositing 500...");
81     al.deposit(500);
82     al.print();
83
84     // Deposit a negative amount - should be rejected
85     write_line("Depositing -200...");
86     al.deposit(-200);
87     al.print();
88
89     // Withdraw a positive amount - should work
90     write_line("Withdrawing 300...");
91     al.withdraw(300);
92     al.print();
93
94     // Withdraw a negative amount - should be rejected
95     write_line("Withdrawing -100...");
96     al.withdraw(-100);
97     al.print();
98
99     // Use get_balance to read the value directly
100    write_line("Current balance (cents): " + to_string(al.get_balance()));
101
102    return 0;
103 }

```



```

8_MemberFunc > C: 8_account.cpp > main()
You, 4 minutes ago | 1 author (You)
1 #include "splashkit.h"
2 #include "utilities.h"
3
4 class account
5 {
6     // The balance of account in cent
7     int balance();
8
9 public:
10    // The name of account holder
11    string name();
12
13    // constructor with param
14    account(string name_, int balance_)
15    {
16        // initialize the account
17        name = name_;
18    }
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1000  // ...

```

- Complete one of the [Test Your Capabilities](#) tasks from the chapter to see that you can now put these tools together to achieve something yourself.

> I choose User account.

First need to think the structure about the user class.

In private I put password, in public I put username, test_login(), change_password().

Test login using bool to check the name and password is both correct.

The condition of change password is old password must correct, old and new password cannot be same, new password again must be correct.

The next step is write main() with while and switch for each case.

>

```

8_memberFunc > C: 8_user.cpp > % user
1 // 4 minutes ago | 1 author (You)
2 #include "splashkit.h"
3 #include "utilities.h"
4
5 class user
6 {
7     string password();
8
9 public:
10     string username();
11
12     // initialize account with param
13     user(string username_, string password_)
14     {
15         username = username_;
16         password = password_;
17     }
18
19     // initialize account without param
20     user()
21     {
22         username = "No name";
23         password = "";
24     }
25
26     // test login, login success when name and password is correct
27     bool test_login(string username_, string password_)
28     {
29         return (username == username_) && (password == password_);
30     }
31
32     // change password
33     // old password must be correct
34     // old and new password cannot be same
35     // new password again must correct
36     bool change_password(string old_password_, string new_password_)
37     {
38         // when current password is wrong
39         if (old_password_ != password)
40         {
41             write_line("Failed to change password - invalid password");
42             return false;
43         }
44         // password reuse
45         else if (old_password_ == new_password_)
46         {
47             write_line("Failed to change password - password reuse");
48             return false;
49         }
50         // password changed, return true
51         else if (old_password_ == password && old_password_ != new_password_)
52         {
53             password = new_password_;
54         }
55     }
56
57     // Backup return
58     return old_password_ == password && old_password_ != new_password_;
59 }
60
61 int main()
62 {
63     user u();
64
65     int option();
66
67     while (option != 4)
68     {
69         // print the menu to user
70         write_line("1: Set Account");
71         write_line("2: Test Login");
72         write_line("3: Change Password");
73         write_line("4: Quit");
74
75         // read option from user
76         option = read_integer("Enter your option: ", 1, 4);
77
78         // doing each case with option
79         switch (option)
80         {
81             // Set account
82             case 1:
83                 // add {} to create a new space
84                 {
85                     // ask username and password
86                     string u_name = read_string("Username: ");
87                     string u_pass = read_string("Password: ");
88
89                     // set account
90                     u = user(u_name, u_pass);
91                     write_line("Account Set");
92
93                     break;
94                 }
95             // test login
96             case 2:
97                 {
98                     // ask username and password
99                     string u_name = read_string("Username: ");
100                     string u_pass = read_string("Password: ");
101
102                     // check whether login succeeded or failed
103                     if (u.test_login(u_name, u_pass))
104                     {
105                         write_line("Login Succeeded");
106                     }
107                     else
108                     {
109                         write_line("Login Failed");
110                     }
111                     break;
112                 }
113             // change password
114             case 3:
115                 {
116                     string old_pass = read_string("Enter current password: ");
117                     string new_pass = read_string("Enter new password: ");
118
119                     u.change_password(old_pass, new_pass);
120
121                     break;
122                 }
123             case 4:
124                 break;
125             default:
126                 break;
127         }
128     }
129
130     return 0;
131 }
132
  
```

```

1 #include "splashkit.h"
2 #include "utilities.h"
3
4 class user
5 {
6     (base) alston@Alstons-MacBook-Air 8_MemberFunc % ./account
7     Current balance (cents): 1200
8     (base) alston@Alstons-MacBook-Air 8_MemberFunc % clang++ 8_user.cpp utilities.cpp -o user -l splashkit -Wall
9     (base) alston@Alstons-MacBook-Air 8_MemberFunc % clang++ 8_user.cpp utilities.cpp -o user -l splashkit -Wall
10    (base) alston@Alstons-MacBook-Air 8_MemberFunc % ./user
11    1: Set Account
12    2: Test Login
13    3: Change Password
14    4: Quit
15    Enter your option: 1
16    Username: Alston
17    Password: 123abc
18    Account Set
19    1: Set Account
20    2: Test Login
21    3: Change Password
22    4: Quit
23    Enter your option: 2
24    Username: Alston
25    Password: 123abc
26    Login Succeeded
27    1: Set Account
28    2: Test Login
29    3: Change Password
30    4: Quit
31    Enter your option: 3
32    Enter current password: 123abc
33    Enter new password: 123abc
34    Failed to change password - password reuse
35    1: Set Account
36    2: Test Login
37    3: Change Password
38    4: Quit
39    Enter your option: 3
40    Enter current password: 1234
41    Enter new password: 123
42    Failed to change password - Invalid password
43    1: Set Account
44    2: Test Login
45    3: Change Password
46    4: Quit
47    Enter your option: 3
48    Enter current password: 123abc
49    Enter new password: 1234
50    Password changed
51    1: Set Account
52    2: Test Login
53    3: Change Password
54    4: Quit
55    Enter your option: 2
56    Username: Alston
57    Password: 1234
58    Login Succeeded

```

Reflect:

- What was the most interesting thing you learnt through this task, and why?

> I learned about the class and the method. Class can encapsulation something to private or public, using class is similar with using the library in python. Also, class have a more complete rule compare with struct, for example is construction function.

- How do you think objects change the way you approach programming?

> I need to think more about the structure of the program I want to write, give it a proper structure using class, struct and function. I use struct to arrange the data, using class to combine the object, data and function with same behaviour. We also can protect the data in class.

- Looking back, how long did it take to complete this task? What did you spend the most time on?

> I use 2 hours to finish this task. I spend time to do the user account system.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I did not use any AI for this task.

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack:
- Engage with feedback and discussion to get this signed off.